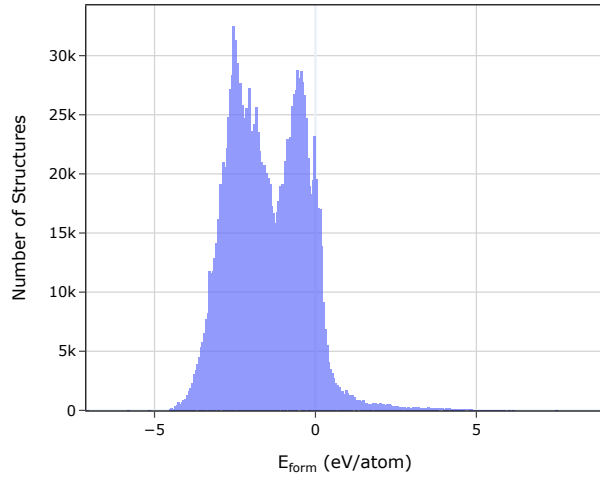
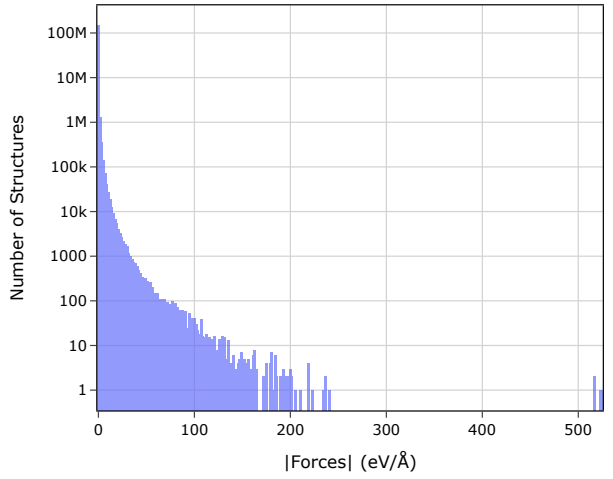
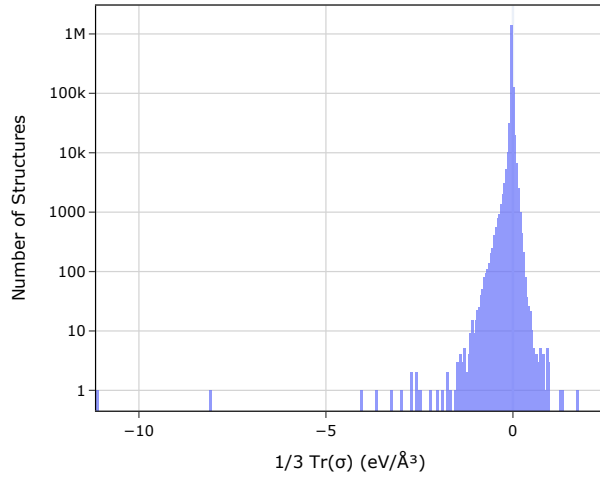




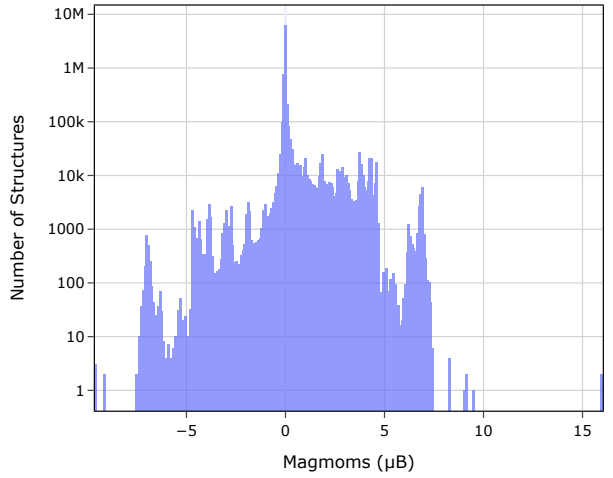
(a) Distribution of formation energies.



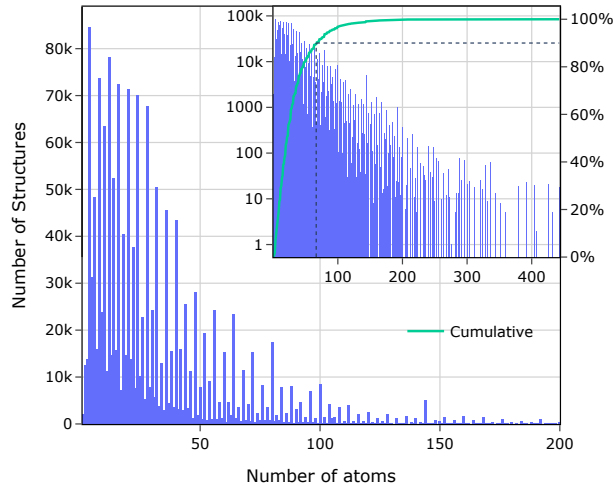
(b) Distribution of forces.



(c) Distribution of stresses.

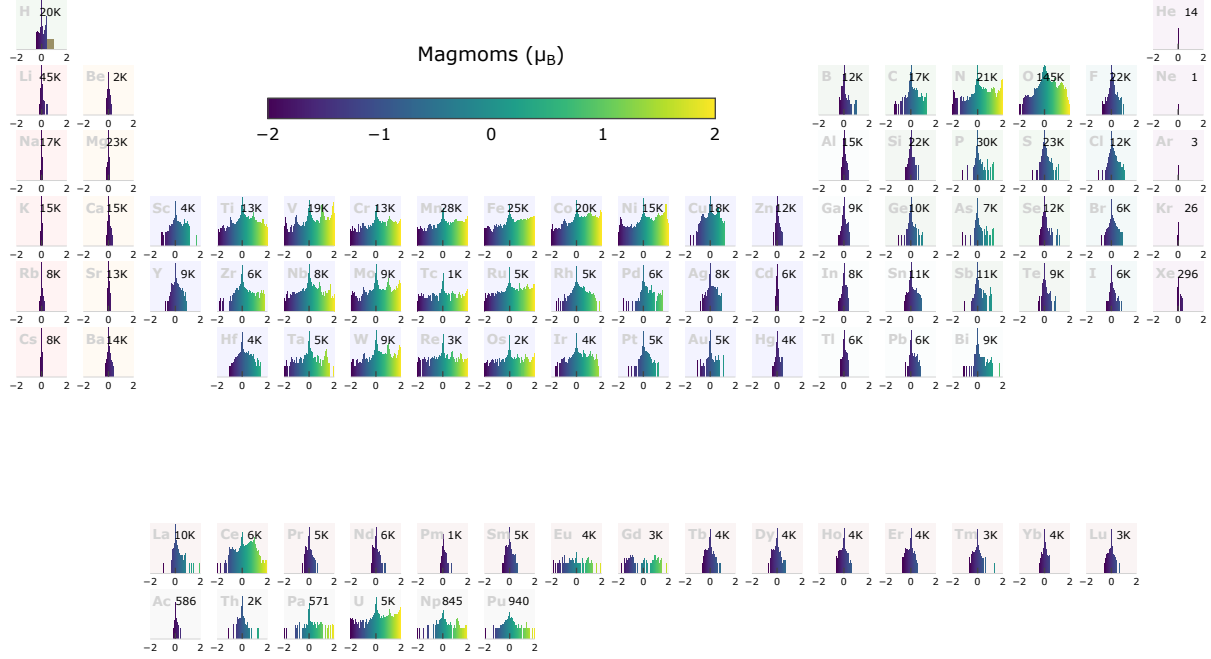


(d) Distribution of magnetic moments.

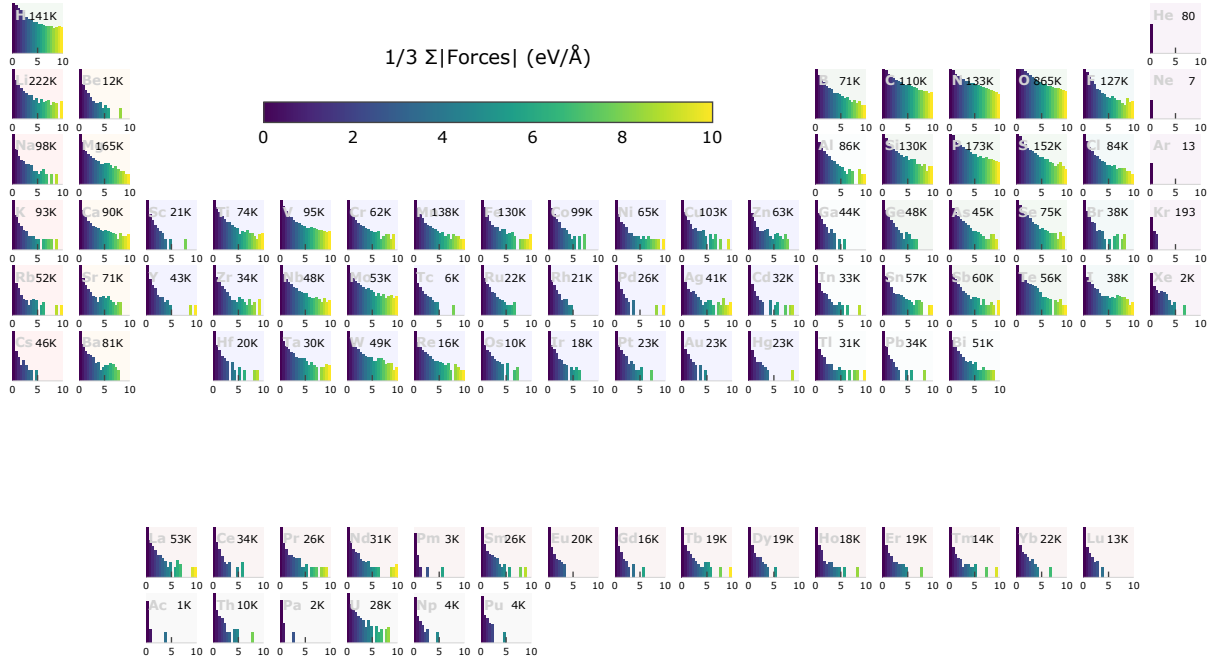


(e) Distribution of the number of atomic sites.

Figure S12: Distribution of energies, forces, stresses, magnetic moments, and number of atoms in MPtrj. Comparing the distribution of formation energies to that of the MP dataset we see that the relative heights of the bimodal peaks is shifted as longer relaxation trajectories are seen on average for materials whose energies are adjusted by the anion correction scheme. The inset in (e) uses a log-scale to show the tail of the distribution. The green cumulative line in the inset shows that 82% have less than 50 sites and 97% of structures in MPtrj have less than 100 atoms.



(a) Element-wise MPtrj magnetic moment distributions



(b) Element-wise MPtrj force distributions

Figure S13: Distribution of magnetic moments and forces for each element MPtrj. This data is used as training targets for all interatomic potentials in this work (only CHGNet uses the absolute value of magnetic moments as targets). The number in the top right corner of each element tile counts the number of target values for that element in all of MPtrj. y -axes are log-scaled to reveal the tail of high magnetic moments in some elements. a) reveals rare erroneous data points in MPtrj. For instance, Cr has a single-point calculation with a highly unphysical magnetic moment of $17\mu_B$. For visualization purposes, the y -axes are again log-scaled and distributions are truncated at $10\text{ eV}/\text{\AA}$. Oxygen has the largest outliers with mean absolute forces of up to $160\text{ eV}/\text{\AA}$.